



'simpler model' = a lower-fidelity model vs the reference — cheetah: paraxial chromatic-linear quad; elegant: trace-space slopes + achromatic drift; impactz: paraxial chromatic thick quad (transfmapK, K/(1+delta))
lighter bar behind + lighter value above = fast-math (relaxed FP), shown when >=5% faster than the IEEE bar

versions: impactx 26.08 · cheetah 0.8.2 · pyat 0.8.0 · pyorbit 3.1-11-gab6bc4b · xsuite 0.53.4 · scibmad v0.4.1 · bmad 20260617-0-1-g67c5d0c83 · elegant elegant-2026.3.0-52-g8b7ea15 · helix v1.9.0 · synergia v2024.04-209-g795b99f39 · impactz V2.7.7